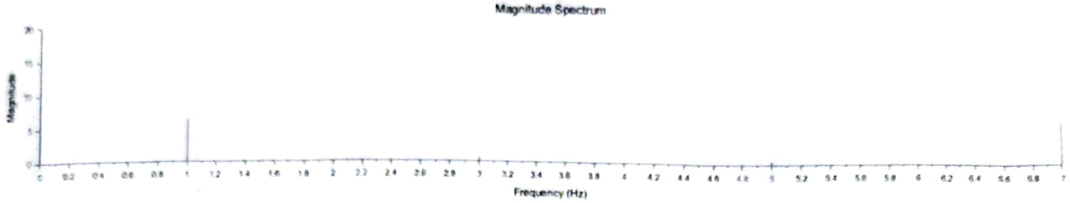
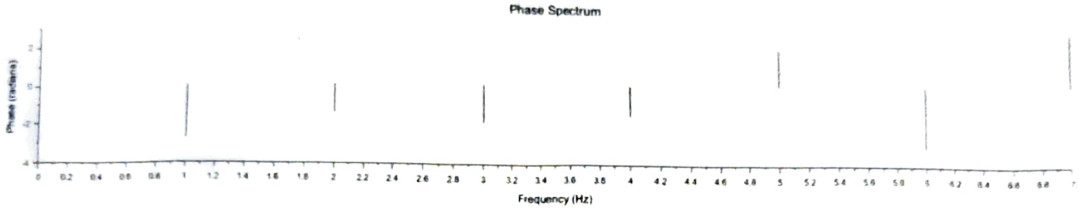

Input Sequence
Amplitude x(n)
Time n


Magnitude Spectrum
Magnitude
Frequency (Hz)


Phase Spectrum
Phase (radians)
Frequency (Hz)

18/10/26

EXP1:- Computational of DFT using DIRECT and FET algorithm without function.

AIM:-

To compute the DFT (Direct fourier transform) of a given discrete - time signal using the direct method and fast fourier Transform (FFT) algorithm without function.

APPARATUS REQUIRED:-

(i) Scilab Software

(ii) Laptop

SOURCE CODE:-

```
clc;
clear;
xn=[1 2 3 4 4 3 2 1];
n1 = 0:1: length(xn)-1;
subplot (3,1,1);
plot2d3 (n1, xn);
xlabel ('Time n');
ylabel ('Amplitude');
title ('Input sequence');
j=sqrt(-1);
N= length (xn);
xk= xeros (1, N); N-1;
for k=0 : 1: length (xk) -1;
for n=0 : N-1;
```

CALCULATION:-

Given:- $x(n) = [1, 2, 3, 4, 4, 3, 2, 1]$

$N = 8$.

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi nk/N}.$$

$$= \sum_{n=0}^{7} x(n) e^{-j\frac{2\pi nk}{8_4}}$$

$$\Rightarrow x(0)e^{0} + x(1)e^{-j\pi k/4} + x(2)e^{-j\frac{2\pi k}{4_2}} + x(3)e^{-j\frac{3\pi k}{4}} + x(4)e^{-j\frac{4\pi k}{4}} + x(5)e^{-j\frac{5\pi k}{4}} + x(6)e^{-j\frac{6\pi k}{4_2}} + x(7)e^{-j\frac{7\pi k}{4}}.$$

$K = 0$,

$= 1+2+3+4+4+3+2+1$

$\Rightarrow 20$

$X(0) = 20$

Magnitude $\Rightarrow \sqrt{a^2+b^2} = \sqrt{400} = 20$

Phase $\Rightarrow \tan^{-1} b/a = 0°$

$K = 1$

$$= 1 + 2\left(\cos\frac{\pi}{4} - j\sin\frac{\pi}{4}\right) + 3\left(\cos\frac{\pi}{2} - j\sin\pi/2\right) + 4\left(\cos\frac{3\pi}{4} - j\sin\frac{3\pi}{4}\right) + 4(\cos\pi - j\sin\pi) + 3\left(\cos\frac{5\pi}{4} - j\sin\frac{5\pi}{4}\right) + 2\left(\cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}\right) + \left(\cos\frac{7\pi}{4} - j\sin\frac{7\pi}{4}\right)$$

$$= 1 + 2(0.707 - j0.707) + 3(0 - j) + 4(-0.707 - j0.707) + 4(-1 - 0) + 3(-0.707 + j0.707) + 2(0 + j) + (0.707 + j0.707).$$

$X(1) = -5.828 - j2.414$

```
xk(k+1) = xk(k+1)+xn(n+1)*exp(-j*2*pi*k
 * n)/N);
 end
 end
disp(xk)
k1 = 0:1:length(xk)-1;
magnitude = abs(xk);
subplot(3,1,2);
plot2d3(k1, magnitude);
xlabel('frequency (Hz)');
Ylabel('magnitude spectrum');
angle = atan(imag(xk), real(xk))
subplot(3,1,3);
plot2d3(k1, angle);
xlabel('frequency (Hz)');
Ylabel('Phase spectrum');
title('Phase spectrum');
```

Magnitude $\Rightarrow \sqrt{(-5.828)^2 + (-2.414)^2}$

≈ 6.308

Phase $\Rightarrow \tan^{-1}\left(\frac{-2.414}{-5.828}\right)$

$= \tan^{-1}(0.414)$

$\theta = 22.5^\circ \Rightarrow -(180^\circ - 22.5^\circ)$

$\Rightarrow -157.5^\circ$

$K = 2,$

$= 1 + 2(\cos\pi/2 - j\sin\pi/2) + 3(\cos\pi - j\sin\pi) + 4\left(\cos\frac{5\pi}{2} - j\sin\frac{5\pi}{2}\right) + 4(\cos 2\pi - j\sin 2\pi) + 3\left(\cos\frac{5\pi}{2} - j\sin\frac{5\pi}{2}\right) + 2(\cos 3\pi - j\sin 3\pi) + \left(\cos\frac{7\pi}{2} - j\sin\frac{7\pi}{2}\right).$

$= 1 + 2(-j) + 3(-1-0) + 4(0+j) + 4(1-0) + 3(0-j) + 2(-1-0) + (0+j)$

$x(2) = 0.$

$K = 3:-$

$= 1 + 2\left(\cos\frac{3\pi}{4} - j\sin\frac{3\pi}{4}\right) + 3\left(\cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}\right) + 4\left(\cos\frac{9\pi}{4} - j\sin\frac{9\pi}{4}\right) + 4(\cos 3\pi - j\sin 3\pi) + 3\left(\cos\frac{15\pi}{4} - j\sin\frac{15\pi}{4}\right) + 2\left(\cos\frac{9\pi}{2} - j\sin\frac{9\pi}{2}\right) + \left(\cos\frac{21\pi}{4} - j\sin\frac{21\pi}{4}\right)$

$= 1+2(-0.707-j0.707)+3(0-j)+4(0.707-j0.707)+4(-0-1)+2(0.707+j0.707)+2(0-j)+1(-0.707+j0.707)$

$X(3) = -0.172 - j2.1716$

Magnitude $\Rightarrow \sqrt{(-0.172)^2+(-2.1716)^2}$

≈ 0.448

Phase $= \tan^{-1}\left(\frac{-2.1716}{-0.172}\right)$

$K=4;$

$X(4) = 1+2e^{-j\pi}+3e^{-j2\pi}+4e^{-j3\pi}+4e^{j4\pi}+3e^{-j5\pi}+2e^{-j6\pi}+e^{j7\pi}$

$= 1+2(-1)+3(1)+4(-1)+4+3(-1)+2-1$

$X(4) = 0$

Magnitude $\Rightarrow \sqrt{(0)^2+(0)^2} = 0$

Phase $\Rightarrow$ Undefined

$K=5;$

$X(5) = 1+2e^{-j5\pi/4}+3e^{-j5\pi/2}+4e^{-j5\pi/4}+4e^{-j5\pi}+3e^{-j\frac{25\pi}{4}}+2e^{-j\frac{15\pi}{2}}+e^{-j\frac{35\pi}{4}}$

$= 1+2(1.0707-j0.707)+3(0+j)+4(0.707-j0.707)+4(-1)+3(0.707+j0.707)+2(0-j)+(-0.707+j0.707)$

$X(5) = -0.172 + j0.414$

Magnitude $\Rightarrow \sqrt{(0.172)^2 + (0.414)^2}$

≈ 0.448 u

Phase $\Rightarrow \tan^{-1}\left(\frac{0.414}{0.172}\right) = 112.5^\circ$

K=6;

$X(6) = 1 + 2e^{-j\frac{3\pi}{2}} + 3e^{-j3\pi} + 4e^{-j9\pi/2} + 4e^{-j6\pi} + 3e^{-j15\pi/2} + 2e^{-j9\pi} + e^{-j21\pi/2}$

$= 1 + 2(j) + 3(-1)(+ 4(-j) + 4(1) + 3(j) + 2(-1) + (-j)$

$X(6) = 0$

Magnitude $\Rightarrow 0$

Phase $\Rightarrow 0$ undefined

K=7;

$X(7) = 1 + 2[0.707 + j0.707] + 3[0 + j] + 4[0.707 - j0.707] + 4[-1] + 3[-0.707 + j0.707] + 2[0 - j] + [0.707 + j0.707)$

$= -5.828 + j2.414$

Magnitude $\Rightarrow$ 6.308

Phase $\Rightarrow 157.5^\circ$

RESULT:-

Thus, the DFT of a given discrete time signal using direct method and FFT algorithm without junction is found successfully.